

```
1 class Student {
2     }
3     public void setName(String name) { // no usage
4         this.name = name;
5     }
6     public void displayProfile() { // usage
7         System.out.println("Name: " + name);
8         System.out.println("Roll Number: " + rollno);
9         System.out.println("Faculty: " + faculty);
10    }
11    public static void main(String[] args) {
12        Student s = new Student("Aava", 02, "BCS");
13        s.displayProfile();
14    }
15 }
```

Run Student

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=61770" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Name: Aava
Roll Number: 2
Faculty: BCS
Process finished with exit code 0
```

30:28 LF UTF-8 4 spaces

classmate

1- Student Information System

```
class Student {
    private String name;
    private int rollno;
    private String faculty;

    public Student (String name, int rollno, String faculty) {
        this.name = name;
        this.rollno = rollno;
        this.faculty = faculty;
    }

    public String getName () {
        return name;
    }

    public int getRollno () {
        return rollno;
    }

    public String getFaculty () {
        return faculty;
    }

    public void displayProfile () {
        System.out.println("Name: " + name);
        System.out.println("Roll Number: " + rollno);
        System.out.println("Faculty: " + faculty);
    }

    public static void main (String [] args) {
        Student s = new Student ("Aava", 02, "BCS");
        s.displayProfile();
    }
}
```

The screenshot shows the IntelliJ IDEA IDE with a project named 'tutorial_3'. The 'src' directory contains four files: 'Main.java', 'Staff.java', 'Employee.java', and 'AdminStaff.java'. The 'Staff.java' file is open, showing the following code:

```
1 class Employee { 4 usages, 2 inheritors
2     void displayDetails() { 2 usages, 2 overrides
3         System.out.println("Employee Details");
4     }
5 }
6
7 class Teacher extends Employee { 1 usage
8     @Override 2 usages
9     void displayDetails() {
10         System.out.println("Teaching Staff");
11     }
12 }
13
14 class AdminStaff extends Employee { 1 usage
15     @Override 2 usages
16     void displayDetails() {
17         System.out.println("Administrative Staff");
18     }
19 }
20
21 public class Staff {
22     public static void main(String[] args) {
23         Employee e1 = new Teacher();
24         Employee e2 = new AdminStaff();
25         e1.displayDetails();
26     }
27 }
```

The bottom of the IDE shows the 'Run' tab with the command: `"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=52912" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8`. The output shows 'Teaching Staff' and 'Administrative Staff'. The status bar at the bottom indicates '24:29 CRLF UTF-8 4 spaces'.

classmate
Date _____
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2. University employee (Teaching or Administrative)

```
class Employee {
    void displayDetails() {
        System.out.println("Employee Details");
    }
}

class Teacher extends Employee {
    void displayDetails() {
        System.out.println("Teaching Staff");
    }
}

class AdminStaff extends Employee {
    void displayDetails() {
        System.out.println("Administrative Staff");
    }
}

public class Main {
    public static void main (String[] args) {
        Employee e1 = new Teacher();
        Employee e2 = new AdminStaff();

        e1.displayDetails();
        e2.displayDetails();
    }
}
```

```

1 abstract class Scholarship { 4 usages
2     abstract void checkEligibility(); 2 usages 2 implementations
3 }
4
5 class MeritScholarship extends Scholarship { 1 usage
6     @Override 2 usages
7     void checkEligibility() {
8         System.out.println("Eligible for Merit Scholarship");
9     }
10 }
11
12 class NeedBasedScholarship extends Scholarship { 1 usage
13     @Override 2 usages
14     void checkEligibility() {
15         System.out.println("Eligible for Need-Based Scholarship");
16     }
17 }
18
19 public class ScholarshipDemo {
20     public static void main(String[] args) {
21         Scholarship merit = new MeritScholarship();
22         Scholarship need = new NeedBasedScholarship();
23     }
24 }

```

Run ScholarshipDemo

```

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=61680" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Eligible for Merit Scholarship
Eligible for Need-Based Scholarship
Process finished with exit code 0

```

3. Student Course Registration

```

import java.util.ArrayList;

class Course {
    String courseName;

    Course(String courseName) {
        this.courseName = courseName;
    }
}

class Student {
    String name;
    ArrayList<Course> courses = new ArrayList<>();

    Student(String name) {
        this.name = name;
    }

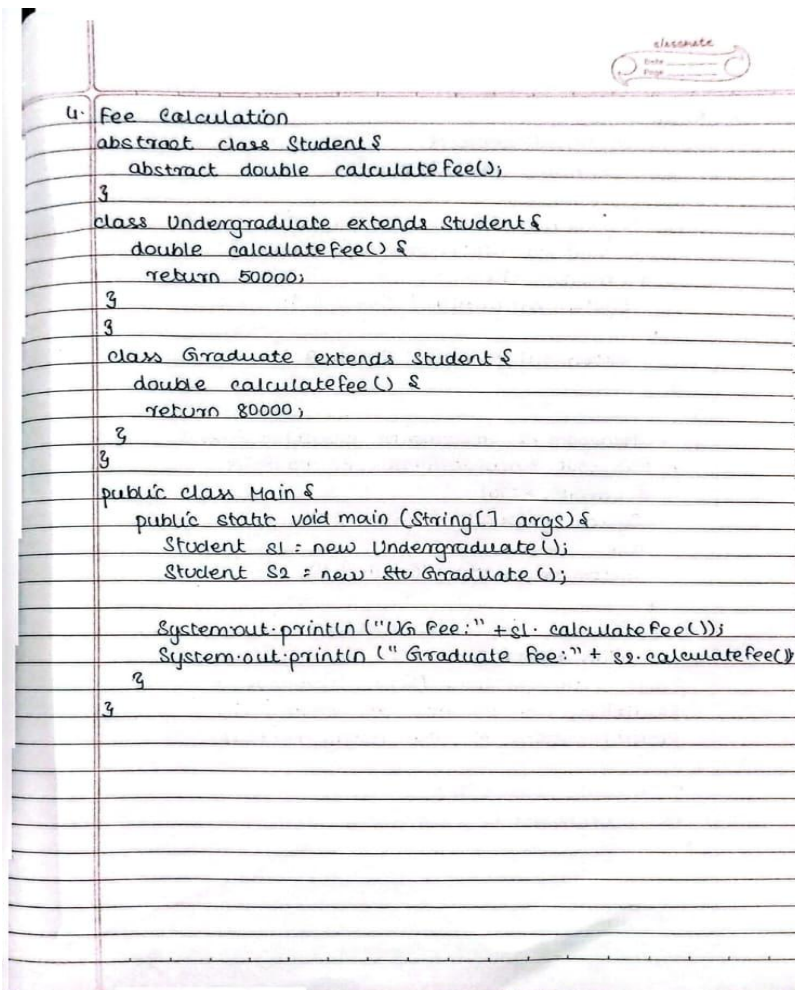
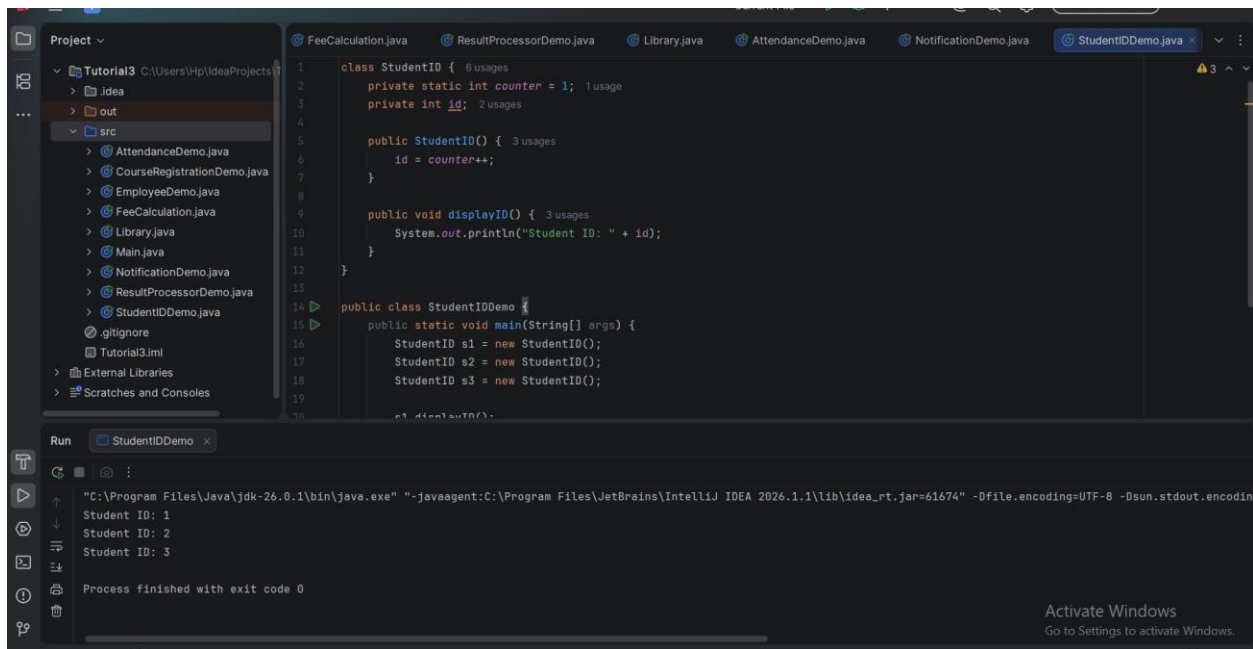
    void addCourse(Course c) {
        courses.add(c);
    }

    void displayCourses() {
        System.out.println("Course registered by " + name + ":");
        for (Course c : courses) {
            System.out.println(c.courseName);
        }
    }

    public static void main(String[] args) {
        Student s = new Student("Aava");
        s.addCourse(new Course("OOP"));
        s.addCourse(new Course("AI"));

        s.displayCourses();
    }
}

```



```
1 interface Notification {
2     void send();
3 }
4
5 class EmailNotification implements Notification {
6     @Override
7     public void send() {
8         System.out.println("Email Notification Sent");
9     }
10 }
11
12 class SMSNotification implements Notification {
13     @Override
14     public void send() {
15         System.out.println("SMS Notification Sent");
16     }
17 }
18
19 public class NotificationDemo {
20     public static void main(String[] args) {
21         EmailNotification email = new EmailNotification();
22         SMSNotification sms = new SMSNotification();
23         email.send();
24         sms.send();
25     }
26 }
```

Run NotificationDemo

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=61659" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8

Email Notification Sent

SMS Notification Sent

Process finished with exit code 0

5 Result

```
interface ResultProcessor {
    void calculateGrade (int marks);
}

class Engineering implements ResultProcessor {
    public void calculateGrade (int marks) {
        if (marks >= 80)
            System.out.println ("Grade A");
        else
            System.out.println ("Grade B");
    }
}

class Management implements ResultProcessor {
    public void calculateGrade (int marks) {
        if (marks >= 70)
            System.out.println ("Grade A");
        else
            System.out.println ("Grade B");
    }
}

public class Main {
    public static void main (String [] args) {
        ResultProcessor e = new Engineering();
        ResultProcessor m = new Management();

        e.calculateGrade (75);
        m.calculateGrade (75);
    }
}
```

```

class Attendance {
    public void calculateAttendance() {
        System.out.println("General Attendance");
    }
}

class EngineeringAttendance extends Attendance {
    @Override
    public void calculateAttendance() {
        System.out.println("Engineering Attendance = 80%");
    }
}

class MedicalAttendance extends Attendance {
    @Override
    public void calculateAttendance() {
        System.out.println("Medical Attendance = 90%");
    }
}

```

Run AttendanceDemo

```

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=51169" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Engineering Attendance = 80%
Medical Attendance = 90%
Process finished with exit code 0

```

classmate

6.

```

class Member {
    int getBorrowLimit() {
        return 0;
    }
}

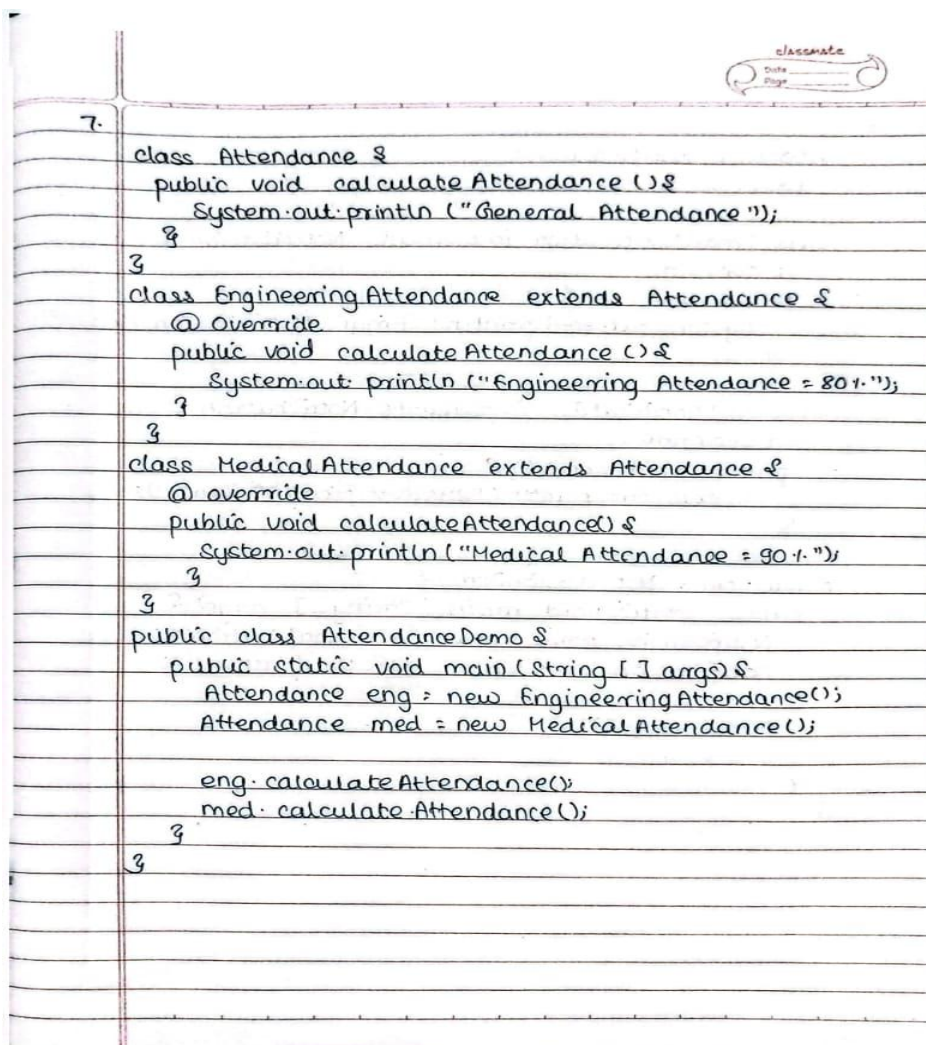
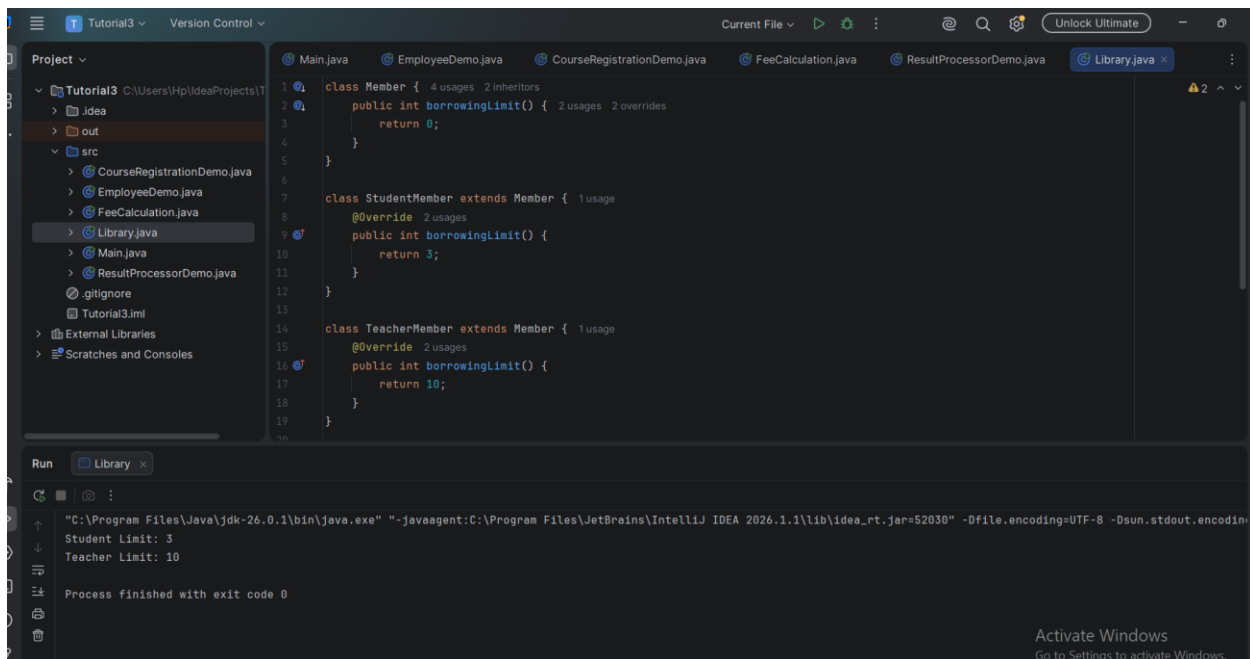
class StudentMember extends Member {
    @Override
    int getBorrowLimit() {
        return 3;
    }
}

class TeacherMember extends Member {
    @Override
    int getBorrowLimit() {
        return 10;
    }
}

public class Librarian {
    public static void main (String [] args) {
        Member s = new StudentMember();
        Member t = new TeacherMember();

        System.out.println("Student limit: " + s.getBorrowLimit());
        System.out.println("Teacher limit: " + t.getBorrowLimit());
    }
}

```



```
1 interface ResultProcessor {
2     void calculateGrade(int marks);
3 }
4
5 class EngineeringDepartment implements ResultProcessor {
6     @Override
7     public void calculateGrade(int marks) {
8         if (marks >= 80)
9             System.out.println("Engineering Grade: A");
10        else
11            System.out.println("Engineering Grade: B");
12    }
13 }
14
15 class ManagementDepartment implements ResultProcessor {
16     @Override
17     public void calculateGrade(int marks) {
18         if (marks >= 70)
19             System.out.println("Management Grade: A");
20        else
21            System.out.println("Management Grade: B");
22    }
23 }
```

Run ResultProcessorDemo

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=63882" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8

Engineering Grade: B
Management Grade: A

Process finished with exit code 0

8.

```
interface Notification {
    void send();
}

class EmailNotification implements Notification {
    @Override
    public void send() {
        System.out.println("Email Notification Sent");
    }
}

class SMSNotification implements Notification {
    @Override
    public void send() {
        System.out.println("SMS Notification Sent");
    }
}

public class NotificationDemo {
    public static void main (String[] args) {
        Notification email = new EmailNotification();
        Notification sms = new SMSNotification();

        email.send();
        sms.send();
    }
}
```

The screenshot shows the IntelliJ IDEA IDE with a project named 'Tutorial3'. The 'Project' view on the left shows the file structure: 'src' contains 'CourseRegistrationDemo.java', 'EmployeeDemo.java', 'FeeCalculation.java', and 'Main.java'. The 'Run' view at the bottom shows the execution of 'FeeCalculation.java', displaying the output: 'Undergraduate Fee = Rs. 50000' and 'Graduate Fee = Rs. 80000'. The 'Main.java' file is open in the editor, showing the following code:

```
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27
```

```
class Graduate extends StudentFee {  
    @Override  
    void calculateFee() {  
        System.out.println("Graduate Fee = Rs. 80000");  
    }  
}  
  
public class FeeCalculation {  
    public static void main(String[] args) {  
        StudentFee ug = new Undergraduate();  
        StudentFee grad = new Graduate();  
        ug.calculateFee();  
        grad.calculateFee();  
    }  
}
```

classmate

9.

```
class StudentID {  
    public static int  
    private int id;  
  
    public StudentID() {  
        id = counter++;  
    }  
  
    public void DisplayID() {  
        System.out.println("Student ID: " + id);  
    }  
}  
  
public class StudentIDDemo {  
    public static void main (String[] args) {  
        Student s1 = new StudentID();  
        Student s2 = new StudentID();  
        Student s3 = new StudentID();  
  
        s1.displayID();  
        s2.displayID();  
        s3.displayID();  
    }  
}
```

